**HEALTH PRODUCTS AND FOOD BRANCH**

**OTTAWA**

# DETECTION AND ISOLATION OF SHIGA TOXIN PRODUCING *ESCHERICHIA COLI* IN FOODS

**Alex Gill[1*], Tanis McMahon[1], Mylène Deschênes[2], Daniel Plante[3] Kyle Tapp[2], and Burton Blais[2]**

**[1] Bureau of Microbial Hazards, Health Canada, 251 Sir Frederick Banting Driveway, Ottawa, Ontario, K1A 0K9**

**[2] Research and Development Section, Ottawa Laboratory Canadian Food Inspection Agency, Ottawa, Ontario, K1A 0C6**

**[3] Microbiology Laboratory Longueuil, Health Canada, 1001 St-Laurent Ouest, Longueuil, Quebec, J4K 1C7**

*** Principal contact:**
**alex.gill@hc-sc.gc.ca**

DRAFT: text of published version may differ.

## 1. Application

This method is applicable to the isolation and identification of Shiga toxin producing *Escherichia coli* (STEC) (also known as verotoxigenic *E. coli*), including enterohemorrhagic *E. coli*, in the food categories Raw Meat Products, Fruit and Vegetable Based Products, and Dairy Products. It can be used to determine compliance with Sections 4 and 7 of the Food and Drug Act.

STEC are a group of pathogenic *E. coli* which are defined by the potential to produce one or more Shiga toxins (Stx, verotoxin) as defined by production of Stx, or presence of the gene *stx*. STEC have multiple evolutionary origins among *E. coli*, and so they are diverse in serotype, genotype, and physiology. STEC infections can result in illness ranging from uncomplicated diarrhea, to bloody diarrhea, and life threatening haemolytic uremic syndrome. STEC possessing specific virulence factors associated with host colonisation have a greater association with severe illness. However, STEC without these factors may still

cause severe illness in vulnerable individuals. This method is intended to support the analysis of food samples for a wide range of STEC allowing the detection and isolation of *E. coli* strains carrying known *stx* variants (1a, 1c, 1d, 1e, 2a to 2o).

## 2. Description

Food samples are stomached in Tryptic Soy broth with pyruvate and incubated at 42 °C for 3h, selective agents vancomycin, cefsulodin and bile salts are then added and incubation continued at 42 °C for a total of 16 to 24 hours. The enrichment broth is screened by real-time PCR for *stx,* using primers and probes that detect 19 toxin subtypes (STX real-time PCR). Enrichment broth cultures positive for *stx* are plated onto both Rainbow agar O157 and MacConkey agar with tellurite and incubated at 35 °C for 16 to 24 hours. Colonies are screened batchwise by the STX real-time PCR to identify potential STEC isolates. Isolates positive for *stx* are characterised with a second real-time PCR for the presence of genes involved in host colonisation (CF real-time PCR). If required further characterisation including species identity by biochemical tests is conducted.

## 3. Principle

STEC in the analytical unit are amplified by incubation in enrichment broth. Enrichment broth potentially containing STEC are identified by detection of *stx* genes with the STX real-time PCR. Isolates are recovered by plating enrichment media onto agar media and screening individual colonies with the STX and CF real-time PCRs targeting *stx* genes and host colonisation factors (*eae, aggR, aaiC*), respectively. Isolates positive for *stx* and any of the colonisation factors are reported as STEC, as these genes have not been reported present together in other species, with exception of strains of the emerging pathogen *Escherichia albertii*. Confirmed positive STEC isolates should also undergo more detailed identification of the virulence profile by STEC CHAS analysis or whole genome sequencing. Since *stx* can be carried by other species of bacteria, *stx*-positive isolates negative for colonisation factors have their species determined by biochemical assay before they can be confirmed STEC.

DRAFT: text of unpublished version not after

## 4. Definition of Terms

See Appendix A of Volume 1.

## 5. Collection of Samples

See Appendix B of Volume 1.

## 6. Materials and Special Equipment

**Note:** The laboratory Supervisor must ensure that completion of the analysis described in this method is done in accordance with the International Standards reference "ISO/IEC 17025:2005 (or latest version): General Requirements for the Competence of Testing and Calibration Laboratories".

The following equipment, materials, media and reagents are required to operate the method and are commercially available. See also Section 11 of this method for the preparation of individual reagents.

**Note:** It is the responsibility of the laboratory to ensure equivalency if any variations of the media formulations listed here are used (either product that is commercially available or made from scratch). Please forward equivalency data to the Editor of *Compendium of Analytical Methods* for consideration of modification of this method.

### 6.1 Equipment

1) Biological safety cabinet (BSC)
2) Refrigerator capable of maintaining 4 °C
3) Freezer capable of maintaining -20 °C
4) Incubators capable of maintaining 35 °C and 42 °C
5) Heat block or water bath capable of accommodating 1.5 mL microcentrifuge tubes and maintaining a temperature of 56, 80 and 100 °C
6) Stomacher, filtered stomacher bags
7) Vortex mixer
8) Microcentrifuge, capable of spinning 1.5 mL to 2 mL tubes at 16,000 x g
9) Micropipettors, adjustable to cover range of volumes from 1.0 to 1000 µL
10) Real-time PCR Thermal cycler with FAM and HEX channels*

DRAFT: text of published version may differ

***Note:** This method has been validated for Roche LightCycler® 96, Bio-Rad CFX96™ and Applied Biosystems™ QuantStudio™ 5 and 6 real-time PCR thermal cycler models using Roche FastStart™ Essential DNA Probes Master buffer. The laboratory can use any equivalent real-time thermal cycler model with compatibility for 6-FAM (520 nm) and HEX (555 nm) dye channels. The laboratory can use any real-time PCR reagents, if equivalence can be demonstrated by performance testing with suitable positive and negative controls.

### 6.2 Materials

1) Gloves, disposable
2) Ice and bucket
3) Screw-cap and regular 1.5 mL microcentrifuge tubes, nuclease-free, sterile
4) Regular 2.0 mL microcentrifuge tubes, nuclease-free, sterile
5) Microcentrifuge tube racks
6) Culture loops, sterile
7) Petri dishes, sterile
8) Pipettes, 10 mL
9) Pipettor tips, filtered, nuclease-free, sterile
10) Test tubes, sterile (10 mL capacity)
11) 96 well plates or 8 tube strips compatible with the Real-Time PCR platform to be used.
12) Optically clear film for 96 well plates

### 6.3 Media and Reagents – Enrichment and strain isolation

1) Cefsulodin sodium salt hydrate (CAS No. 52152-93-9)
2) Vancomycin hydrochloride (CAS No. 1404-93-9)
3) Bile salts no. 3
4) Sodium pyruvate (CAS No. 113-24-6)
5) InstaGene Matrix (Bio-Rad, Cat. no. 732-6030)
6) Tween 20 (CAS No. 9005-64-5)
7) Puregene Cell Kit (8 x $10^8$) (Qiagen, Cat. no. 158043)
8) Isopropanol (CAS No. 67-63-0)
9) Ethanol (CAS No. 64-17-5)
10) 0.1% (w/v) Peptone water
11) Sodium chloride (NaCl) (CAS No. 7647-14-5)
12) Sodium Phosphate, monobasic, monohydrate ($NaH_2PO_4.H_2O$) (CAS No. 10049-21-5)
13) Sodium phosphate, dibasic, anhydrous ($Na_2HPO_4$) (CAS No. 7558-79-4)

DRAFT: text of published verison may differ

14) Potassium phosphate, dibasic ($K_2HPO_4.H_2O$) (CAS No. 7758-11-4)
15) Tryptic Soy broth (TSB)
16) Brain Heart Infusion (BHI) agar or Nutrient agar (NA)
17) MacConkey (MAC) agar
18) Potassium Tellurite (CAS No. 7790-58-1)
19) Rainbow® Agar O157 (RBA) dehydrated media (Biolog Inc., Cat. no. 80102)

### 6.4 Media and Reagents – Template preparation and real-time PCR

1) DNase and RNase free water (Ultrapure or molecular grade)
2) 1X TE (Tris-EDTA buffer pH 8, 10 mM Tris, 1 mM EDTA)
3) FastStart Essential DNA Probes Master mastermix (Roche Diagnostics, Cat. no. 06 402 682 001)
4) Triton X-100 (CAS No. 9036-19-5)
5) Oligonucleotides for Stx, CF and IPC – See section 10 and 11 for specifics on primers, probes, and master mix
6) IPC template (4 nmol of dried Duplex Ultramer DNA) – refer to section 10 and 11

### 6.5 Control cultures

Positive controls:
A combination of *E. coli* strains that cover at least one Shiga toxin gene (*stx1* and/or *stx2*) and at least one host colonisation factor (*eae and/or aggR and/or aaiC*).

Negative control:
Generic *E. coli* negative for all virulence characteristics. No Shiga toxin genes (*stx1* or *stx2*) and no host colonisation factors (*eae, aggR, aaiC*).

**Note:** It is the responsibility of each laboratory to ensure that the temperatures of the incubators or water baths are maintained at the recommended temperatures. Where 35 °C or lower is recommended in text of the method the incubator may be at 35 °C ± 1.0 °C. However, where higher temperatures are recommended, such as 42 °C, it is imperative that the incubators or water baths be maintained within 0.5 °C due to potential lethality of the higher temperatures on the microorganism(s) being isolated.).

DRAFT: text of published verison may differ

## 7. Procedure

The test shall be carried out in accordance with the following instructions:

### 7.1 Handling of Sample Units

7.1.1 Upon the receipt of samples at the analytical laboratory, shelf-stable foods may be stored at room temperature, refrigerated foods should be stored between 1 to 6 °C, and frozen foods should be stored between -15 to -30 °C. Thaw frozen foods in a refrigerator between 1 to 6 °C.

7.1.2 Analyze sample units as soon as possible after their receipt in the laboratory.

### 7.2 Preparation for analysis

7.2.1 Have sterile Tryptic Soy broth with pyruvate (TSBp) (9 mL per g of analyte) prepared and pre-warmed to 42 °C ± 0.5 °C before use.

7.2.2 Clean the surface of the working area with a suitable disinfectant.

### 7.3 Preparation of sample

7.3.1 To ensure a representative analytical unit, agitate liquids or free flowing materials until the contents are homogeneous. If the sample unit is a solid, obtain the analytical unit by taking a portion from several locations within the sample unit.

7.3.2 Place the analytical unit in a stomacher bag and prepare a 1 in 10 dilution of the food by aseptically adding the enrichment broth, TSBp. The recommended minimum analytical unit is 25 g or 25 mL diluted into 225 mL of TSBp. STEC may be present in outbreak associated foods at levels below 1 cell per 25 g and so larger analytical unit sizes, such as 65 g or mL in 585 mL or 100 g or mL in 900 mL may be used while maintaining the 1 in 10 dilution. For raw meats analytical units up to 325 g in 2925 mL of TSBp are permitted. Do not exceed the sample size of 325 g.

7.3.3 Stomach the sample for 2 minutes.

DRAFT: Text of published version may differ

## 7.4 Enrichment

7.4.1 Incubate the homogenized sample without shaking for 3 hours at 42 °C ± 0.5 °C.

7.4.2 Prepare selective agents for addition, mix an equal volume of 1 mg/mL vancomycin + 0.3 mg/mL cefsulodin solution with 150 g/L bile salts no. 3 solution in a sterile container.

7.4.3 Following the 3 hours of enrichment, for each 100 mL of enrichment add 2 mL of the mixed selective agents (i.e. 25 g in 225 mL TSBp, add 5 mL of selective agents solution). This gives a final concentration in the enrichment broth of 10 µg/mL vancomycin, 3 µg/mL cefsulodin, and 1.5 g/L bile salts no.3. Working in a biological safety cabinet, add the selective agents to the sample and mix manually for 10 seconds by gently squeezing the bag.

7.4.4 Continue the enrichment by incubation overnight for a total of 16-24 hours at 42 °C ± 0.5 °C without shaking.

7.4.5 Set up a positive and a negative control at the same time (Section 6.5 for details).

## 7.5 DNA Extraction from Enrichment Broth

7.5.1 Following completion of the overnight incubation, screen the enrichment broth for the presence of Shiga toxin genes. To avoid working with a large volume of enrichment broth, use a sterile pipet to transfer a minimum of 10 mL of broth into a labelled sterile tube. Keep the remainder of the enrichment broth between 1 and 7 °C, until the results of the screening STX real-time PCR (Section 7.6) are known.

**Note:** This method has been validated for two methods for the extraction of DNA from enrichment broths, Bio-Rad InstaGene Matrix and Qiagen Puregene Cell Kit. The Qiagen Puregene Cell Kit is specifically recommended for Nut and Nut Meats, or if the presence of DNA inhibitors in the food matrix is detected or suspected (McMahon et al. 2023).

7.5.2 **DNA template preparation from enrichment broths using InstaGene Matrix**

DRAFT: text of published version may differ

7.5.2.1 Transfer a 1.5 mL aliquot of the enrichment broth into a sterile 2 mL microcentrifuge tube.

7.5.2.2 Centrifuge at 500 x g for 1 minute. Transfer a 1 mL aliquot of the supernatant into a sterile 1.5 mL centrifuge tube.

7.5.2.3 Centrifuge the tube at 4 °C for 3 minutes at 10,000 x g.

7.5.2.4 Remove the supernatant from the tube and re-suspend the pellet in 1 mL of sterile, 1% (v/v) Tween 20 solution. Set up a lysis blank negative control by adding 1 mL of sterile, 1% (v/v) Tween 20 solution to a sterile 1.5 mL centrifuge tube and carrying it through the rest of the procedure. Vortex at high speed for 10 seconds.

7.5.2.5 Centrifuge the tube at 4 °C for 3 minutes at 10,000 x g.

7.5.2.6 Remove and discard the supernatant from the pellet. Mix InstaGene Matrix on a magnetic stirrer to maintain the matrix in suspension (the product bottle comes with a stir bar). Add 200 µL of InstaGene Matrix to the pellet (use a large bore pipet tip, such as a 1000 µL tip) and vortex for at least 10 seconds.

7.5.2.7 Incubate the tube with InstaGene Matrix at 56 °C for 15 to 30 minutes.

7.5.2.8 Mix the contents by vortexing for 10 seconds and then incubate the tube at 100 °C for 8 minutes.

7.5.2.9 Cool the tube on ice for 5 minutes, vortex the tube for 10 seconds, then centrifuge at 4 °C for 3 minutes at 10,000 x g.

7.5.2.10 Retain 50 µL of the supernatant as template for the screening STX real-time PCR (step 7.6.4).

7.5.3 **DNA template preparation from enrichment broths using Puregene Kit.**

7.5.3.1 Transfer a 1.5 mL aliquot of the enrichment broth into a sterile 2 mL microcentrifuge tube.

DRAFT: text of published version may differ

7.5.3.2 Centrifuge at 500 x g for 1 minute.

7.5.3.3 Transfer 1 mL of the supernatant to a 1.5 mL microcentrifuge tube on ice.

7.5.3.4 Centrifuge for 5 seconds at 16,000 x g to pellet cells (let the centrifuge reach the speed before starting the 5 seconds). Carefully discard the supernatant by pipetting.

7.5.3.5 Add 300 µL of Cell Lysis Solution, and vortex to mix (ensure the pellet is completely suspended). Set up a lysis blank negative control by adding 300 µL Cell Lysis Solution to a sterile 1.5 mL centrifuge tube and carrying it through the rest of the procedure. Incubate sample at 80°C for 5 minutes to lyse the cells and then incubate for 1 minute on ice to quickly cool the sample.

**Note:** Samples are stable in Cell Lysis Solution for at least 2 years at room temperature (15–25°C).

7.5.3.6 Add 100 µL of Protein Precipitation Solution, and vortex vigorously for 20 seconds at high speed. Incubate on ice for 5 minutes. Centrifuge for 3 minutes at 16,000 x g.

7.5.3.7 Pipet 300 µL of isopropanol into a new 1.5 mL microcentrifuge tube. Then, leaving the protein pellet in the first tube, transfer the supernatant from the previous step by pipette to the isopropanol and invert 50 times to mix.

7.5.3.8 Centrifuge for 1 minute at 16,000 x g. The DNA will be a small white pellet though it is possible a pellet will not be visible.

7.5.3.9 Carefully discard the supernatant by pipetting (leave some liquid at bottom to avoid aspirating the pellet), and drain the open tube by inverting on a clean piece of absorbent paper, taking care that the pellet remains in the tube.

**Note:** Multiple samples can be inverted by placing tubes in a foam rack (or equivalent holder), placing absorbent paper on top and inverting foam rack and absorbent paper together.

7.5.3.10 Add 300 µL of 70% ethanol and invert several times to wash the DNA. Centrifuge for 1 minute at 16,000 x g. Carefully discard the supernatant by pipetting (leave some liquid at bottom to avoid aspirating the pellet). Drain the open tube by inverting on a clean

DRAFT: text of published version may differ

piece of absorbent paper, taking care that the pellet remains in the tube. Allow the open tube to air dry for 5 minutes.

7.5.3.11 Add 50 µL of DNA Hydration Solution and vortex to dissolve the pellet. This extracted DNA will be used as template for broth screening STX real-time PCR (step 7.6.4).

## 7.6 Enrichment screening with STX real-time PCR

7.6.1 Label reaction tubes or 96-well plates.

7.6.2 Prepare the STX real-time PCR Master Mix for the number of reactions needed. For master mix recipes, primer sequences, probe sequences, preparation of working stocks, and storage of oligos, refer to Section 10.

7.6.3 Before dispensing to tubes or wells of 96-well plates, add 1 µL per reaction of Internal Positive Control (IPC) dsDNA template to the master mix (e.g. 10 µL IPC template for 10 reactions). Mix and dispense 22.5 µL into reaction tubes or wells of 96-well plates adapted to the real-time PCR thermal cycler used.

7.6.4 Add 2.5 µL of extracted DNA sample template (step 7.5.2.10 or 7.5.3.11) to corresponding tubes or wells (including positive and negative controls).

7.6.5 Briefly Spin down PCR strips or plate to collect the liquid at the bottom of the tube. Insert tubes or 96-well plate into a real-time PCR thermal cycler and proceed with the PCR amplification using the program described in Section 10.8. The test is considered positive for *stx* if the FAM channel exceeds the fluorescence signal of negative controls within 35 cycles. The test is considered negative for *stx* if the FAM channel does not exceed the fluorescence signal of negative controls, and the HEX (IPC) channel produces a detectable signal within 35 cycles. Refer to Section 10.8 for an example of a typical amplification curve for a *stx* positive sample.

Failure to produce a fluorescence signal in the FAM and HEX channel for a sample indicates a failed reaction, probably due to inhibition of the PCR by sample matrix contaminants. The test is considered invalid and should be repeated with DNA extract diluted by a factor of 10 and 100 in molecular grade water.

DRAFT: text of published version may differ

Failure to produce a fluorescence signal in the FAM channel for the positive control indicates omission of a critical component in the procedure (e.g., PCR reaction mixture). The test is considered invalid and should be repeated with newly prepared reagents.

**Table 1.** STX real-time PCR results legend

| FAM s*tx* (+/-) | HEX IPC (+/-) | STX real-time PCR Results |
|---|---|---|
| + | + | Positive |
| + | - | Positive[a] |
| - | + | Negative |
| - | - | Invalid |

[a] IPC in HEX channel can be negative from reagent depletion by *stx* amplification

7.6.6. Enrichment broths testing positive for *stx* are plated on Rainbow agar O157 (RBA) and MacConkey agar with 2.5 µg/mL tellurite (MAC-T) to isolate STEC as described in Section 7.7.1.

7.6.7. If enrichment broths test negative for *stx,* end analysis and report the sample as “STEC not detected”. Refer to Section 8 ‘Reporting Results’.

## 7.7 Strain Isolation

7.7.1 Prepare serial dilutions of the enrichment broth to $10^{-6}$ by mixing 1 mL aliquots of the enrichment broth with 9 mL of 0.1% peptone water.

7.7.2 Spread-plate 100 µL of dilutions of the enrichment media onto RBA and MAC-T, plating dilutions that ideally produce 50-100 well separated colonies per plate. At a minimum, plate dilutions 10-4, 10-5 and 10-6 onto RBA, and dilutions 10-1, 10-2 and 10-3 onto MAC-T. Other dilutions may be plated as necessary to provide sufficient separated colonies.

7.7.3 Incubate plates for 16 to 24 hours at 35 °C ± 1 °C.

7.7.4 For *stx*-positive enrichment broths, select and pool isolated colonies to screen by STX real-time PCR for any *stx*. Select colonies at random, ignoring pinprick size colonies and bearing in mind the highly variable characteristics of STEC. STEC O157 colonies on RBA are black or grey, but other STEC may form colonies of a wide range of colours. The

DRAFT Text of published version may differ

majority of *E. coli* strains form pink/red colonies on MAC-T, a minority of strains do not utilize lactose and form translucent colonies. It is important to pick random pools of colonies from the agar plates, so that all the various phenotypes are included in the pool. For batch-wise colony screening, pick colonies from the plate dilution(s) showing the largest number of individual colonies, ideally, 50-100 colonies per plate. Start with 2 pools of 10 colonies, 1 pool from RBA and 1 pool from MAC-T. If no colonies are observed on both RBA and MAC-T plated with the lowest dilution of enrichment broth, the sample is considered negative.

7.7.5 Pick a portion of a single colony with a loop, touch the loop onto a non-selective agar plate such as BHI, or nutrient agar (labelled with a grid to keep track of individual colonies).

7.7.6 Suspend the remainder of the colony in 100 µL of 1% (v/v) Triton X-100 in a sterile 1.5 mL microcentrifuge tube. Repeat the procedure, and suspend the remaining 9 colonies that constitute a pool into the same microcentrifuge tube.

7.7.7 Incubate the non-selective plate for 16-24 hours at 35 °C ± 1 °C.

7.7.8 Prepare the template for STX real-time PCR by vortexing the tubes containing the colony pool for 10 seconds, and then place the tubes into a 100 °C heating block for 10 minutes. Cool the resulting lysates for 5 minutes at room temperature and pulse-spin for a few seconds to bring down condensation accumulated under the tube cap. Use 2.5 µL of the supernatant as the PCR template.

7.7.9 Test each colony pool for the presence of *stx* using the STX real-time PCR protocol in Section 10.8. For *stx*-positive results continue to 7.7.10. In the absence of any *stx*-positive results from a colony pool, pick another 2 pools of 10 colonies each and process in like manner. Repeat additional rounds if necessary, for a minimum of 60 colonies. If a minimum of 60 colonies are tested for *stx* and none are positive, end analysis and report result as “STEC not detected.” Refer to Section 8 ‘Reporting Results’.

7.7.10 To identify STEC from positive colony pools, go back to the isolated colonies from that pool (on non-selective agar plate) and test them individually for *stx* by STX real-time PCR. Prepare colony lysate as in step 7.7.6. In the instance of time-sensitive investigations (e.g. foodborne

DRAFT: text of published version may differ

illness outbreaks), test colonies from *stx*-positive colony pools individually for *stx* by STX real-time PCR and for colonisation factors by CF real-time PCR. Combining both STX and CF real-time PCRs together (in the same 96-well plate but different wells) will save overall time of analysis.

Alternatively, colonies may be identified using the STEC CHAS (MFLP – 119). Lysates from colony isolates may be stored at 4 °C and used as template for the STEC-CHAS characterisation.

7.7.11 Any colonies testing positive for *stx* with the STX real-time PCR are considered presumptive STEC isolates and must be confirmed as STEC isolates by following the steps outlined below (7.7.12 to 7.7.13).

7.7.12 To identify STEC from positive colonies, use the positive lysate from the STX real-time PCR analysis or go back to the positive isolated colony from the positive pool (on non-selective agar plate) and test them individually for colonisation factors by CF real-time PCR. Prepare individual colony lysates as in step 7.7.6 (or use lysates from 7.7.10) and perform analysis of these lysates by following steps 7.6.2 to 7.6.5, but use the CF master mix (Section 10.7) to replace STX master mix.

7.7.13 If a presumptive STEC isolate is positive for colonisation factors by CF real-time PCR or positive for one or more of the individual colonisation factors *eae*, *aagR*, or *aaiC* by STEC CHAS, report the sample as “STEC isolated” (See Section 8 Reporting Results).

If a presumptive STEC isolate is negative for colonisation factors, determine the species identity of the isolate by VITEK, API, or equivalent method. If the isolate is confirmed as *E. coli* and is *stx* positive, report the sample as “STEC isolated” (See Section 8 ‘Reporting Results’). If the isolate is not confirmed as *E. coli* and negative for colonisation factors report the sample as “STEC not detected” (See Section 8 ‘Reporting Results’).

STEC isolates can be characterized further by STEC CHAS (MFLP – 119) or whole genome sequencing.

DRAFT: text of published version may differ

**Table 2.** Colony screening real-time PCR results legend

| STX (+/-) | CF (+/-) | Species Identity | Result (+/-) |
|---|---|---|---|
| STX (+) | CF (+) | NA | Positive STEC (+) |
| STX (+) | CF (-) | *E. coli* | Positive STEC (+) |
| STX (+) | CF (-) | Not *E. coli* | Negative STEC (-) |
| STX (-) | CF (+) | NA | Negative STEC (-) |
| STX (-) | CF (-) | NA | Negative STEC (-) |

NA: Not Applicable

DRAFT: text of published verison may differ

## 8. Reporting Results

**A.** If the enrichment is negative for *stx* by STX real-time PCR, end analysis and report as **"STEC not detected in analytical unit of ___ g or mL"**

**B.** If a minimum of 60 colonies isolated from the enrichment are negative for *stx* by STX real-time PCR, end analysis and report as **"STEC not detected in analytical unit of ___ g or mL"**

**C.** If one or more colony isolates are positive for *stx* by STX real-time PCR or STEC CHAS and positive for colonisation factors by CF real-time PCR or STEC CHAS, end analysis and report as **"STEC isolated from analytical unit of ___ g or mL".**

**D**. If one or more colony isolates are positive for *stx* by STX real-time PCR or STEC CHAS, negative for colonisation factors by CF real-time PCR or STEC CHAS, and are confirmed *Escherichia coli* by VITEK or API (or equivalent method), end analysis and report as **"STEC isolated from analytical unit of ___ g or mL".**

**E**. If one or more colony isolates are positive for *stx* by STX real-time PCR or STEC CHAS, negative for colonisation factors by CF real-time PCR or STEC CHAS, and are not confirmed *Escherichia coli* by VITEK or API (or equivalent method), end analysis and report as **"STEC not detected in analytical unit of ___ g or mL".**

DRAFT: text of published version may differ

## 9. References


9.1 Boisen, N., Melton-Celsa, A. R., Scheutz, F., O'Brien, A. D., and Nataro, J. P. (2015). Shiga toxin 2a and Enteroaggregative *Escherichia coli*--a deadly combination. Gut microbes, 6(4): 272–278, DOI: 10.1080/19490976.2015.1054591

9.2 EU Reference Laboratory for *Escherichia coli* including Verotoxigenic *E. coli* (EURL-VTEC) (2021). Identification and characterization of Shiga Toxin (Verocytotoxin)-producing *Escherichia coli* (STEC/VTEC) by Real Time PCR amplification of the main virulence genes and the genes associated with the serogroups mainly associated with severe human infections. Department of Food Safety, Nutrition and Veterinary Public Health, Unit of Food Microbiology and Foodborne Diseases, Istituto Superiore di Sanità, EURL-VTEC_Method_02_Rev 1. https://www.iss.it/documents/20126/0/EURL-VTEC_Method_02_Rev_1.pdf

9.3 EURL-VTEC (2021). Detection of Enteroaggregative *Escherichia coli* in food by Real Time PCR amplification of the *aggR* and *aaiC* genes. Department of Food Safety, Nutrition and Veterinary Public Health, Unit of Food Microbiology and Foodborne Diseases, Istituto Superiore di Sanità, EURL-VTEC_Method_05_Rev 2. https://www.iss.it/documents/20126/0/EURL-VTEC_Method_05_Rev+2.pdf

9.4 Huszczynski, G., Gauthier, M., Mohajer, S., Gill, A., and Blais. B. (2013). Method for the detection of priority Shiga toxin-producing *Escherichia coli* in beef trim. Journal of Food Protection, 76 (10):1689-1696. DOI: 10.4315/0362-028X.JFP-13-059

9.5 Gill, A., Martinez-Perez, A., McIlwham, S., and Blais, B. (2012). Development of a method for the detection of verotoxin-producing *Escherichia coli* in food. Journal of Food Protection, 75:827-837. DOI: 10.4315/0362-028X.JFP-11-395

9.6 Gill, A., Huszczynski, G., Gauthier, M., and Blais, B. (2014). Evaluation of eight agar media for the isolation of Shiga toxin–producing *Escherichia coli.* Journal of Microbiological Methods. 96:6-11. DOI: 10.1016/j.mimet.2013.10.022

9.7 Gill, A., Dussault, F., McMahon, T., Petronella, N., Wang, X., Cebelinski, E., Scheutz, F., Weedmark, K., Blais, B., and Carrillo, C. (2022). Characterization of atypical Shiga toxin gene sequences and description of Stx2j, a new subtype. Journal of Clinical Microbiology, 60(3): e0222921. DOI: 10.1128/jcm.02229-21


DRAFT: text of published version may differ

9.8 Gomes, T., Ooka, T., Hernandes, R. T., Yamamoto, D., and Hayashi, T. (2020). *Escherichia albertii* pathogenesis. EcoSal Plus, 9(1): 10.1128/ecosalplus.ESP-0015-2019. DOI: 10.1128/ecosalplus.ESP-0015-2019

9.9 Kavlick, M.F. (2018). Development of a universal internal positive control. Biotechniques, 65(5): 275-280. DOI: 10.2144/btn-2018-0034

9.10 McMahon, T., Abdelmesih, M., and Gill, A. (2023). Evaluation of DNA extraction methods for the detection of Shiga toxin producing *Escherichia coli* in food by polymerase chain reaction, International Journal of Food Microbiology, 404:110317. DOI: 10.1016/j.ijfoodmicro.2023.110317

9.11 McMahon T, Clarke S, Deschênes M, Tapp K, Blais B, Gill A. (2024). Real-time PCR primers and probes for the detection of Shiga toxin genes, including novel subtypes. Int J Food Microbiol. 419:110744. DOI: 10.1016/j.ijfoodmicro.2024.110744.

9.13 Scheutz, F., Teel, L.D., Beutin, L., Piérard, D., Buvens, G., Karch, H., Mellmann, A., Caprioli, A., Tozzoli, R., Morabito, S., Strockbine, N.A., Melton-Celsa, A.R., Sanchez, M., Persson, S., and O'Brien, A.D. (2012). Multicenter evaluation of a sequence-based protocol for subtyping Shiga toxins and standardizing Stx nomenclature. Journal of Clinical Microbiology, 50(9): 2951-2963. DOI: 10.1128/JCM.00860-12

9.14 Wang X, Yu D, Chui L, Zhou T, Feng Y, Cao Y, Zhi S. (2024). A comprehensive review on Shiga toxin subtypes and their niche-related distribution characteristics in Shiga-toxin-producing *E. coli* and other bacterial hosts. Microorganisms. 12(4):687. DOI: 10.3390/microorganisms12040687

DRAFT: text in published version may differ

# APPENDICES

## 10. Oligonucleotide sequences, master mix recipes, thermal cycling programs

### 10.1 STX real-time PCR Oligonucleotide Primer Sequences

| Primer name | Sequence (5' à 3') | Amplicon size (bp) | Targeted *stx* subtypes[a] |
|---|---|---|---|
| vtxI-F | TTATCGCTTTGCTGATTTTTCAC | 166 | 1a, 1c, 1d, 1e |
| vtxII-R | ATACGACATCAAATCCAGATAAGA | | 1d, 1e |
| vtxIII-R | ATGCGACATTAAATCCAGATAAGA | | 1a, 1c |
| vtxII-F | TACAGATTTTCGGATTTTGCG | 169 | 2j |
| vtxIII-F | TTACAGATTCTCCGATTTCTCAC | | 2f |
| vtxIV-F | CTACAGGTTTTCAGATTTTGCAC | | 2b, 2c, 2d, 2e, 2h, 2i, 2k, 2l, 2n, 2o |
| vtxV-F | CTACCGTTTTTCAGATTTTACAC | | 2a, 2b, 2c, 2g, 2m |
| vtxI-R | CTCCATTAAATCCAGATATGAACC | | 2f |
| vtxIV-R | CTCCATTAACGCCAGATATGATG | | 2a, 2b, 2c, 2d, 2e, 2g, 2h, 2i, 2j, 2k, 2l, 2m, 2n, 2o |
| IPC-F[b] | CGCGAGATACACTGCCAGAA | 65 | |
| IPC-R[b] | GACCACAGCCAGATTAAATTTACCA | | |

[a] Primers for *stx* subtypes were designed from various sources:

- Primers targeting *stx* subtypes 1a, 1c, 1d, 1e, 2a, 2b, 2c, 2d, 2e, 2f, 2g were designed from sequences compiled in Scheutz *et al.*, 2012
- Primers targeting *stx* subtypes 2h, 2i, 2k, 2l were designed from GenBank accession no. CP022279, FN252457, CP041435, AM904726, respectively.
- Primers targeting *stx* subtypes 2j, 2o were designed from sequences described in Gill *et al.,* 2022

[b] IPC real-time PCR primers designed by Kavlick, 2018.

DRAFT: Draft unpublished verison may differ

### 10.2 STX real-time PCR Oligonucleotide Probe Sequences

| Probe name | Fluorophore Quencher combination[a] | Sequence (5' à 3') | Targeted *stx* subtypes[h] |
|---|---|---|---|
| vtxI-P[b] | 5' 6-FAM / ZEN / 3' IBFQ | TACAAGAGCGGTTACATTGTCTGGT GACAGTA | 1a, 1d, 1e |
| vtxII-P[c] | 5' 6-FAM / ZEN / 3' IBFQ | TGTGACAACTATTTCCATGACAACG GACAGCA | 2e, 2f, 2h, 2i, 2k, 2l |
| vtxIII-P[d] | 5' 6-FAM / ZEN / 3' IBFQ | TGTGACAACGGTTTCCATGACAACG GACAGCA | 2a, 2b, 2c, 2d, 2g, 2j, 2m, 2n, 2o |
| vtxIV-P[e] | 5' 6-FAM / ZEN / 3' IBFQ | TACAACTGCGGTTACATTGTCTGGT GACAGTA | 1c |
| IPC-P[f] | 5' HEX / 3' MGBEc[g] | TCCGCGTGATTACG | |

[a] Refer to Section 11.2.2 for instructions on ordering real-time PCR probes
[b] vtxI-P real-time PCR probe (32-mer, 44% G+C)
[c] vtxII-P real-time PCR probe (32-mer, 44% G+C)
[d] vtxIII-P real-time PCR probe (32-mer, 50% G+C)
[e] vtxIV-P real-time PCR probe (32-mer, 44% G+C)
[f] IPC-P real-time PCR probe (14-mer, 57% G+C, designed from Kavlick, 2018)
[g] Minor Groove Binding probe (MGBEc)
[h] Probes for *stx* subtypes were designed from various sources:

- Probes targeting *stx* subtypes 1a, 1c, 1d, 1e, 2a, 2b, 2c, 2d, 2e, 2f, 2g were designed from sequences compiled in Scheutz *et al.*, 2012
- Probes targeting *stx* subtypes 2h, 2i, 2k, 2l were designed from GenBank accession no. CP022279, FN252457, CP041435, AM904726, respectively.
- Probes targeting *stx* subtypes 2j, 2o were designed from sequences described in Gill *et al.,* 2022

DRAFT: text of published version may differ

### 10.3 CF real-time PCR Oligonucleotide Primer Sequences

| Primer name | Sequence (5' à 3') | Amplicon size (bp) |
|---|---|---|
| eae-F[a] | CATTGATCAGGATTTTTCTGGTGATA | 102 |
| eae-R[a] | CTCATGCGGAAATAGCCGTTA | |
| aggR-F[b] | GAATCGTCAGCATCAGCTACA | 102 |
| aggR-R[b] | CCTAAAGGATGCCCTGATGA | |
| aaiC-F[c] | CATTTCACGCTTTTTCAGGAAT | 160 |
| aaiC-R[c] | CCTGATTTAGTTGATTCCCTACG | |
| IPC-F[d] | CGCGAGATACACTGCCAGAA | 65 |
| IPC-R[d] | GACCACAGCCAGATTAAATTTACCA | |

[a] eae real-time PCR primers, designed by EURL-VTEC from GenBank accession no. Z11541.
[b] aggR real-time PCR primers, designed by EURL-VTEC from GenBank accession no. CU928159.2.
[c] aaiC real-time PCR primers, designed by EURL-VTEC from GenBank accession no. NC_011748.1.
[d] IPC real-time PCR primers, designed by Kavlick, 2018.

### 10.4 CF real-time PCR Oligonucleotide Probe Sequences

| Probe name | Fluorophore Quencher combination[a] | Sequence (5' à 3') |
|---|---|---|
| eae-P[b] | 5' 6-FAM / ZEN / 3' IBFQ | ATAGTCTCGCCAGTATTCGCCACCAATACC |
| aggR-P[c] | 5' 6-FAM / ZEN / 3' IBFQ | CGGACAACTGCAAGCATCTA |
| aaiC-P[d] | 5' 6-FAM / ZEN / 3' IBFQ | CACATACAAGACCTTCTGGAGAA |
| IPC-P[e] | 5' HEX / 3' MGBEc[f] | TCCGCGTGATTACG |

[a] Refer to Section 11.2.2 for instructions on ordering real-time PCR probes
[b] eae-P real-time PCR probe (30-mer, 50% G+C, designed by EURL-VTEC from GenBank accession no. Z11541)
[c] aggR-P real-time PCR probe (20-mer, 50% G+C, designed by EURL-VTEC from GenBank accession no. CU928159.2)
[d] aaiC-P real-time PCR probe (23-mer, 44% G+C, designed by EURL-VTEC from GenBank accession no. NC_011748.1)
[e] IPC-P real-time PCR probe (14-mer, 57% G+C, designed from Kavlick, 2018)
[f] Minor Groove Binding probe (MGBEc)

DRAFT not published verison may differ

### 10.5 Internal Positive Control (IPC) dsDNA template sequence

| Name | Sequence (5' à 3') |
|---|---|
| IPC dsDNA | CGC GAG ATA CAC TGC CAG AAA TCC GCG TGA TTA CGA GTC GTG GTA AAT TTA ATC TGG CTG TGG TC |
| | GAC CAC AGC CAG ATT AAA TTT ACC ACG ACT CGT AAT CAC GCG GAT TTC TGG CAG TGT ATC TCG CG |

IPC dsDNA template (65-mer, 49% G+C, from Kavlick, 2018).
Synthesize oligo as "Duplexed Ultramer DNA Oligos". Refer to Section 11.2.3 for preparation of IPC dsDNA template.

### 10.6 STX real-time PCR master mix

| Reagent | Volume (µL) | [Final] |
|---|---|---|
| Molecular Grade Water | 0.62 | |
| 2 x FastStart Essential DNA Probes Master | 12.5 | 1 x |
| STX primers pool (10 µM each)[a] | 3.13 | 1.25 |
| STX probes pool (5 µM each)[b] | 1.25 | 0.25 |
| IPC-F (1 µM)[c] | 1.25 | 0.05 µM |
| IPC-R (5 µM)[c] | 1.5 | 0.3 µM |
| IPC-P (5 µM)[c] | 1.25 | 0.25 µM |
| DNA template (lysate) | 2.5 | |
| IPC dsDNA template (6250 copies / µL)[d] | 1 | 250 copies / µL (6250 copies) |
| Total Volume | 25 | |

[a,b,c] Refer to Section 10.1 for primer sequences, to Section 10.2 for probe sequences, and to Section 11.2 for the preparation and storage of oligonucleotide working stocks.
[d] IPC dsDNA template (Section 10.5) must be added alongside template (lysate). In order to avoid contamination of clean PCR reagents, add both DNA templates (sample + IPC dsDNA) *outside* of the master mix preparation area.
Master mix may be prepared in larger quantities, aliquoted in portions into 1.5 mL screw-cap tubes, and stored at -20°C for up to 6 months.

DRAFT: text of published version may differ

### 10.7 CF real-time PCR master mix

| Reagent | Volume (µL) | [Final] |
|---|---|---|
| Molecular Grade Water | 0.62 | |
| 2 x FastStart Essential DNA Probes Master | 12.5 | 1 x |
| CF primers pool (10 µM each)[a] | 3.13 | 1.25 |
| CF probe pool (5 µM each)[b] | 1.25 | 0.25 |
| IPC-F (1 µM)[c] | 1.25 | 0.05 µM |
| IPC-R (5 µM)[c] | 1.5 | 0.3 µM |
| IPC-P (5 µM)[c] | 1.25 | 0.25 µM |
| DNA template (lysate) | 2.5 | |
| IPC dsDNA template (6250 copies / µL)[d] | 1 | 250 copies / µL (6250 copies) |
| Total Volume | 25 | |

[a,b,c] Refer to Section 10.3 for primer sequences, to Section 10.4 for probe sequences, and to Section 11.2 for the preparation and storage of oligonucleotide working stocks.
[d] IPC template (Section 10.5) must be added alongside template (lysate). In order to avoid contamination of clean PCR reagents, add both DNA templates *outside* of the master mix preparation area.
Master mix may be prepared in larger quantities, aliquoted in portions into 1.5 mL screw-cap tubes, and stored at -20°C for up to 6 months.

### 10.8 STX and CF real-time PCR cycling program

| Step | Temperature | Time |
|---|---|---|
| 1. Polymerase activation | 95 °C | 5 minutes |
| 2. 35 cycles of: | | |
| Denaturation | 95 °C | 20 seconds |
| Amplification / Detection | 57 °C | 1 minute |

DRAFT: text of published version differ

**Figure 1:** Example of a *stx*-positive result from the STX real-time PCR with Roche LightCycler 96. FAM channel (*stx*) solid line, HEX channel (IPC) dashed line.

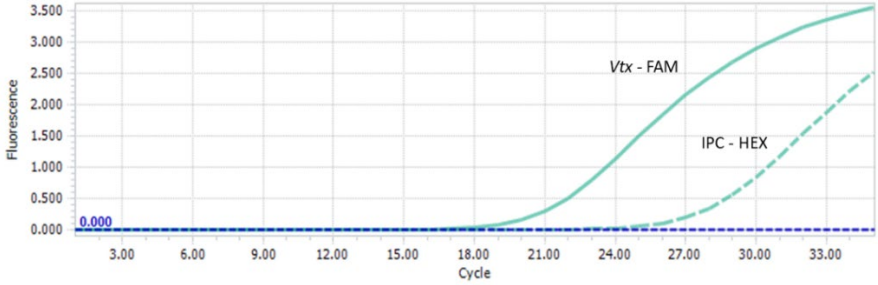



**Figure 2:** Example of a CF-positive result from the CF real-time PCR with Roche LightCycler 96. FAM channel (CF) solid line, HEX channel (IPC) dashed line.

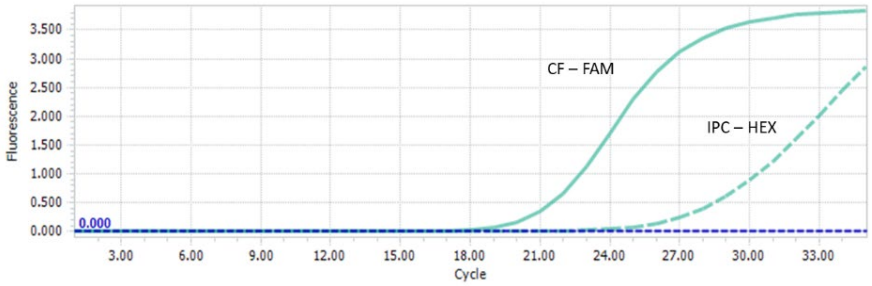
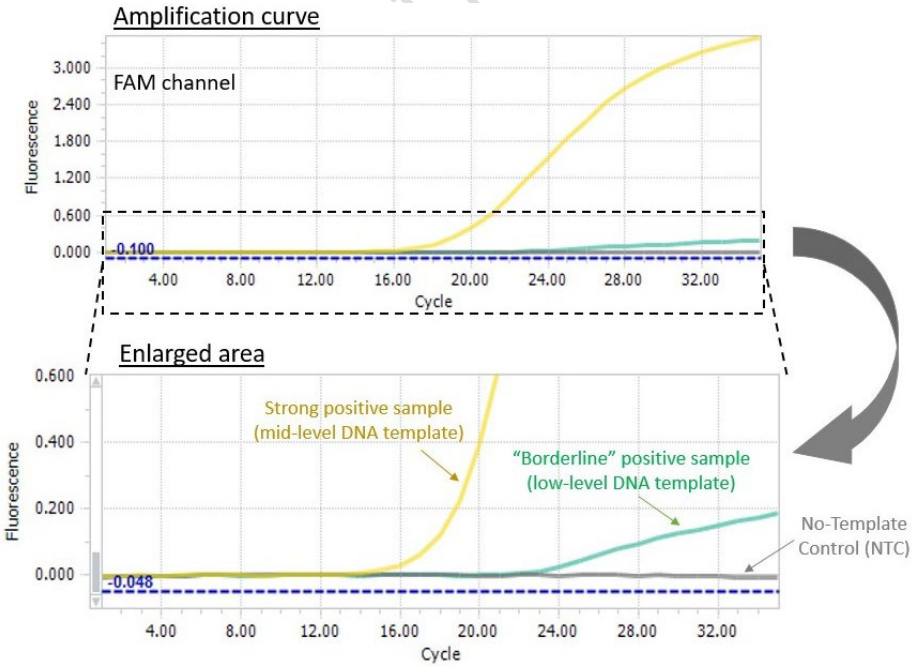



**Figure 3**: Example of a positive "borderline" result with low-level DNA template (green line) compared to a strong positive sample (yellow line) in the FAM channel with Roche LightCyler96. The fluorescence signals of the positive samples increase on the y-axis, relative to the No-Template Control (NTC) which remains flat along the x-axis. In the case of a negative sample, the signal would resemble that of the NTC.



# 11. Media and Reagent Preparation

## 11.1 Bacteriological Media

**Brain Heart Infusion Agar (BHI)**

| Components | Amount (g/L) |
|---|---|
| Brain infusion solids | 12.5 |
| Beef heart infusion solids | 5.0 |
| Protease peptone | 10.0 |
| Sodium chloride | 5.0 |
| Di-sodium phosphate | 2.5 |
| Glucose | 2.0 |
| Agar | 10.0 |

Prepare according to manufacturer's instructions.
After pouring plates, ensure they are thoroughly dried before use. Store at 4-10 °C for up to eight weeks.

**Nutrient Agar**

| Components | Amount (g/L) |
|---|---|
| 'Lab-Lemco' Powder | 1.0 |
| Yeast extract | 2.0 |
| Peptone | 5.0 |
| Sodium chloride | 5.0 |
| Agar | 15.0 |

Prepare according to manufacturer's instructions.
After pouring plates, ensure they are thoroughly dried before use. Store at 4-10 °C for up to eight weeks.

**Tryptic Soya Broth with Pyruvate (TSBp)**

| Components | Amount (g/l) |
|---|---|
| Pancreatic digest of casein | 17.0 |
| Papaic digest of soybean meal | 3.0 |
| Sodium chloride | 5.0 |
| Dipotassium phosphate | 4.0 |
| Glucose | 2.5 |
| Sodium Pyruvate | 1.0 |

Add solid ingredients to 1000 mL distilled water, mix well and sterilize by autoclaving at 121°C for 15 min. Store at 4-10 °C for up to eight weeks.

DRAFT text of published version may differ

**Vancomycin [1 mg/mL] + Cefsulodin [0.3 mg/mL] Stock Solution**

| Components (for 10 ml of solution) | Amount |
|---|---|
| Vancomycin hydrochloride | 10 mg |
| Cefsulodin sodium salt hydrate | 3 mg |

Dissolve the antimicrobials in 10 mL ultrapure water and sterilize the solution by passing through a 0.22 µm filter, into a sterile container such as a Falcon tube. Store at 4 °C for up to 2 weeks.

**Bile Salts No. 3 [150 g/L] Stock Solution**

| Components (for 10 ml of solution) | Amount |
|---|---|
| Bile Salts No. 3 | 1.5 g |

Dissolve the bile salts in 10 mL ultrapure water and sterilize the solution by passing through a 0.22 µm filter, into a sterile container such as a Falcon tube. Store at 4 °C for up to 2 weeks.

**MacConkey Agar (MAC)**

| Components | Amount (g/L) |
|---|---|
| Pancreatic digest of gelatin | 17.0 |
| Peptones (meat and casein) | 3.0 |
| Lactose | 10.0 |
| Bile Salts No. 3 | 1.5 |
| Sodium chloride | 5.0 |
| Agar | 13.5 |
| Neutral Red | 0.03 |
| Crystal Violet | 0.001 |

Prepare according to manufacturer's instructions.
After pouring plates, ensure they are thoroughly dried before use. Store at 4-10 °C for up to six weeks.

**MacConkey Agar with Potassium Tellurite [2.5 mg/L] (MAC-T)**

To make Potassium Tellurite stock solution (1 mg/mL), add 0.1 g Potassium Tellurite salt into 100 mL water. Filter sterilize. Store at 4 °C for up to two weeks.

Prepare MAC according to manufacturer's instructions. After tempering 500 mL media to 45-50 °C, aseptically add 1.25 mL Potassium Tellurite stock solution. Swirl to mix, pour and store plates as described above for MAC.

DRAFT: text of published version may differ

**Rainbow® Agar O157 (RBA) - 500 mL**
Obtained from Biolog Inc., Cat. No. 80102
Rainbow Agar O157 30 g packet (approx. 25 plates)
Prepare according to manufacturer's instructions.
After pouring plates, ensure they are thoroughly dry before use. Store in the dark at 4-10 °C for up to two weeks.

**0.1% Peptone Water – 1 L**
Dissolve 1 g of Proteose Peptone in 1 litre of distilled water. Dispense into desired aliquots and sterilize by autoclaving at 121 °C, 15 psi for 15 min. Store at 4-10 °C for up to six weeks.

**1% Tween 20 – 100 mL**
Dissolve 1 mL Tween 20 in 99 mL ultrapure water (millipore). Filter sterilise and store at room temperature for up to 1 year.

**1% Triton X-100 - 50 mL**
0.5 mL Triton X-100 in 49.5 mL ultrapure water. Filter sterilise. Store at room temperature for up to 1 year.

DRAFT: text of published verison may differ

## 11.2 Oligo preparation and storage

### 11.2.1 Preparation of real-time PCR primers

Resuspend dried oligo in 1X TE (Tris-EDTA buffer pH 8, 10 mM Tris, 1 mM EDTA) at a 100 µM concentration.

To reduce the number of pipetting steps when preparing the STX and/or CF real-time PCR master mix, a pool of the forward and reverse primers may be mixed in equimolar amounts as follows:

**STX primers pool (10 µM each) [a]**

| | Volume (µL) | [Final] |
|---|---|---|
| vtxI-F (100 µM) | 100 | 10 µM |
| vtxI-R (100 µM) | 100 | 10 µM |
| vtxII-F (100 µM) | 100 | 10 µM |
| vtxII-R (100 µM) | 100 | 10 µM |
| vtxIII-F (100 µM) | 100 | 10 µM |
| vtxIII-R (100 µM) | 100 | 10 µM |
| vtxIV-F (100 µM) | 100 | 10 µM |
| vtxIV-R (100 µM) | 100 | 10 µM |
| vtxV-F (100 µM) | 100 | 10 µM |
| 1X TE | 100 | |

**CF primers pool (10 µM each) [a]**

| | Volume (µL) | [Final] |
|---|---|---|
| eae-F (100 µM) | 100 | 10 µM |
| eae-R (100 µM) | 100 | 10 µM |
| aaiC-F (100 µM) | 100 | 10 µM |
| aaiC-R (100 µM) | 100 | 10 µM |
| aggR-F (100 µM) | 100 | 10 µM |
| aggR-R (100 µM) | 100 | 10 µM |
| 1X TE | 400 | |

To avoid repeated freeze/thaw cycles of STX and CF oligos it is advised to prepare aliquots of 10 µM working pools stocks dispensed into screw-cap tubes and store at -20ºC until required.

To prepare a working IPC Forward primer at 1 µM, dilute 10 µL of stock oligo at 100 µM with 990 µL 1 X TE.

DRAFT - Text of published version may differ

To prepare a working IPC Reverse primer at 5 µM, dilute 50 µL of stock oligo at 100 µM with 950 µL 1 X TE.

To avoid repeated freeze/thaw cycles of IPC primers oligos it is advised to prepare aliquots of working stocks dispensed into screw-cap tubes and store at -20ºC until required.

11.2.2 Preparation of real-time PCR probes

When ordering probes (all except IPC), select 5' 6-FAM™ dye / ZEN™ / 3' IB®FQ quencher. Upon arrival, resuspend dried oligo in 1X TE (Tris-EDTA buffer pH 8, 10 mM Tris, 1 mM EDTA) at a 100 µM concentration.

To reduce the number of pipetting steps when preparing the STX and/or CF real-time PCR master mix, a pool of the probes may be mixed in equimolar amounts as follows:

**STX probes pool (5 µM each) [b]**

| | Volume (µL) | [Final] |
|---|---|---|
| vtxI-P (100 µM) | 50 | 5 µM |
| vtxII-P (100 µM) | 50 | 5 µM |
| vtxIII-P (100 µM) | 50 | 5 µM |
| vtxIV-P (100 µM) | 50 | 5 µM |
| 1X TE | 800 | |

**CF probes pool (5 µM each) [b]**

| | Volume (µL) | [Final] |
|---|---|---|
| eae-P (100 µM) | 50 | 5 µM |
| aaiC-P (100 µM) | 50 | 5 µM |
| aggR-P (100 µM) | 50 | 5 µM |
| 1X TE | 850 | |

To avoid repeated freeze/thaw cycles of STX and CF oligos it is advised to prepare aliquots of 5 µM working pools stocks dispensed into screw-cap tubes. Store at -20ºC until required.

For IPC probe only, select 5' HEX™ dye / 3' MGBE quencher. When ordering select 20 or 50 nmol under "Normalized yield (nmol)", 100 µM under "Conc (µM)" and TE Buffer under "Diluent".

DRAFT text of published version may differ

To prepare a working IPC probe at 5 µM, dilute 50 µL of stock oligo at 100 µM with 950 µL1 X TE.

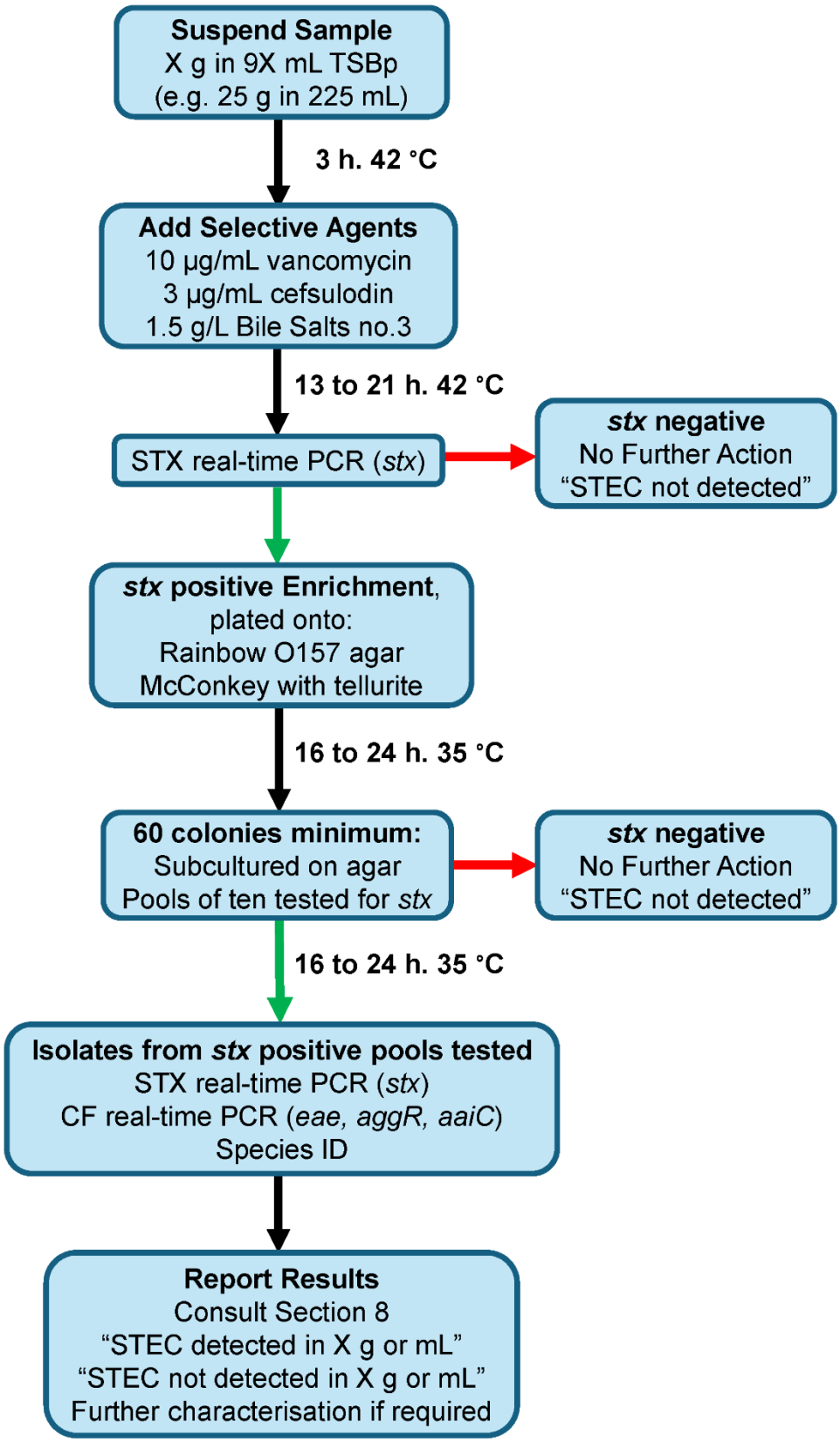
To avoid repeated freeze/thaw cycles of IPC oligos it is advised to prepare aliquots of 5 µM working stocks dispensed into screw-cap tubes. Store at -20ºC until required.

### 11.2.3 Preparation of IPC dsDNA template

Starting with 4 nmol of dried Duplex Ultramer DNA, resuspend in 1 mL of 1X TE (Tris-EDTA buffer pH 8, 10 mM Tris, 1 mM EDTA) to make 2.41 x $10^{12}$ copies / µL. Perform a series of four 1:100 dilutions (10 µL in 990 µL of 1X TE) to make 2.41 x $10^{4}$ copies / µL, then mix 259.3 µL of this dilution with 740.7 µL of 1X TE for a final concentration of 6250 copies / µL IPC dsDNA.

### 11.2.4 Quality Control Verification of STX Primer and Probe Pools

To ensure that no primer or probe was missed when preparing the STX primer pool and STX probe pool it is recommended that a PCR be conducted with these pools, with 19 individual reactions with plasmids representing target sequences of the 19 different *stx*-subtypes (https://github.com/bmh-vteclab/VT-CF_RealTimePCR_Controls/blob/main/vt_realtimePCR_QCcontrols.txt) at 10 pg/µL of template.

DRAFT: text of published version may differ

## 12. Method Workflow